Marmara University
Engineering Faculty
Electrical and Electronics Engineering
EE4077: Fundamentals of Machine Learning
Midterm Exam (Spring, 2023)

1. In this problem, you will see how to learn a binary Naive Bayes classifier. Consider the dataset below. You are looking to buy a car; the label is 1 if you are interested in the car and 0 if you aren't. There are three features: whether the car is red (your favorite color), whether the car is affordable, and whether the car is fuel-efficient.

Interested?	Red?	Affordable?	Fuel-Efficient?
1	1	1	1
0	0	1	0
0	0	1	1
1	0	0	0
0	0	1	1
0	0	1	1
1	1	1	1
1	1	0	1
0	0	0	0

(a) [5 pts] How many parameters do we need to learn?
(b) [5 pts] Estimate the parameters via MLE..
(c) [5 pts] If I see a car that is red, not affordable, and fuel-efficient, would the classifier predict that I would be interested in it?
(d) [5 pts.] Is there a problem with this classifier based on your calculations for the previous question? If so, how can we fix it?

2. In this problem, you will see how to learn a binary logistic regression classifier. Consider the dataset below, where Y is the class label and Xs are attributes

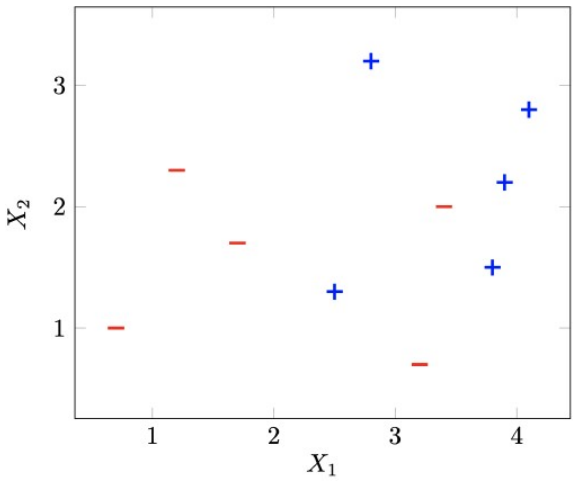
Y	X_1	X_2	X_3
1	1	2	1
1	1	1	-1
0	1	-2	1

(e) [5 pts] What is classification error $J(\theta)$ if the initial weight vector for the attributes is equal to $\theta = [-2,2,1]^T$
(f) [5 pts] Calculate $\frac{\partial J^{(1)}(\theta_1)}{\theta_1}$, $\frac{\partial J^{(1)}(\theta_2)}{\theta_2}$, $\frac{\partial J^{(1)}(\theta_3)}{\theta_3}$ for the first training sample (first row). Note that $\sigma(3) \approx 0.95$
(g) [5 pts] Calculate $\frac{\partial J^{(1)}(\theta_1)}{\theta_1}$, $\frac{\partial J^{(1)}(\theta_2)}{\theta_2}$, $\frac{\partial J^{(1)}(\theta_3)}{\theta_3}$ for the second training sample (first row). Note that $\sigma(-1) \approx 0.25$
(h) [5 pts.] Assuming we are doing stochastic gradient descent with a learning rate of 1.0, what are the updated parameters θ if we update θ using the second training example?
(i) [5 pts] What is the new $J(\theta)$ after doing the above update? Should it decrease or increase?
(j) [5 pts] Given a test example where ($X_1 = 1, X_2 = 3, X_3 = 4$), what will the classifier output following this update?

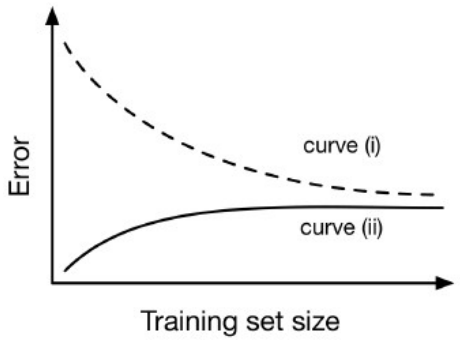
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4. In this problem, we will consider the effect of training dataset size N on a logistic regression classifier with d features. The classifier is trained by optimizing the conditional log-likelihood. The optimization procedure stops if the estimated parameters perfectly classify the training data or they converge. The following plot shows the general trends in training and testing error as we increase the sample size $N = |S|$



(a) [5 pts] Which curve represents the training error? Provide 1-2 sentences of justification.
(b) [5 pts] In one word, what does the gap between the two curves represent?

5. (a) [5 pts] What are the four main tasks in machine learning?
(b) [5 pts] What is the corresponding machine learning task if want to estimate the price of a house based on a dataset. Indicate the underlined machine learning task if want to decide if an email is spam or not based on a dataset.

SOLUTIONS

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a) Tabloya bakmadan = basta araba aldığımızı bilmeli = 1 P(Y)
— kaç parametre varsa o kadar (2 ihtimal)

3 parametre
1 + (2+2+2) = 7

$\frac{P(Y|Y) = 6}{P(Y=1)}$

*Tablada bir sütunun tamamı 1 veya 0 olsaydı bile yine böyle yapardık. 1 + (parametre kadar 2)

b) Parametreleri MLE yönt. bul. MLE (max likelihood estimation) → Tablodaki oranları direkt yaz, sadece.

	Y=1	Y=0
Red	3/4	0
Affordable	1/2	4/5
fuel-efficient	3/4	3/5

c) Araba kırmızı ✓
not aff. ✗
fuel eff. ✓
alırmıyım? P(Y=1) > P(Y=0) mı?

$P(Y=1) \cdot P(Red=1|Y=1) \cdot P(Aff=0|Y=1) \cdot P(fuel\,eff.=1|Y=1) = \frac{4}{9} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{3}{4}$

$P(Y=0) \cdot P(Red=1|Y=0) \cdot P(Aff=0|Y=0) \cdot P(fuel\,eff=1|Y=0) = \frac{5}{9} \cdot 0 \cdot \frac{1}{5} \cdot \frac{3}{5} = 0$

You infrested that car. Because P(Y=1) > P(Y=0)

d) ihtimali = 0 olan biz kesin alırız deriz. ÖRN: P(Y=0 | Red=1) = 0

When car is red, I buy always then.

This a problem. So we can fix this, adding 1 fake (sayı) count, in table

Red dan arabayı almama iht = 0 yani kırmızı olan her arabayı alırım.

P(Y=0 | Aff=0) = 0

Affordable olmayan arabaların hepsini alırım demektir.

logistic regressin sorusu

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	Y	X_1	X_2	X_3
z1	1	1	2	1
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(e) [5 pts] What is classification error $J(\theta)$ if the initial weight vector for the attributes is equal to $\theta=[-2,2,1]^T$

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(h) [5 pts.] Assuming we are doing stochastic gradient descent with a learning rate of 1.0, what are the updated parameters θ if we update θ using the second training example?

(i) [5 pts] What is the new $J(\theta)$ after doing the above update? Should it decrease or increase?

(j) [5 pts] Given a test example where ($X_1 = 1, X_2 = 3, X_3 = 4$), what will the classifier output following this update?

Bu tür sorularda tek bir "z" sihirli sayısı bulup, tüm soruyu onun üzerinden çözeriz.

e) $\theta = [-2,2,1]^T$ (Y_1, Y_2, Y_3)

1.satır $z_1 = (-2\cdot 1) + (2\cdot 2) + (1\cdot 1) = 3$

2.satır $z_2 = -2\cdot 1 + (2\cdot 1) + (1\cdot -1) = -1$

3.satır $z_3 = -2\cdot 1 + (2\cdot -2) + 1\cdot 1 = -5$

Bu son tarzında Y=1 ise $-\log(\sigma(z))$

Y=0 ise $-\log(1-\sigma(z))$

hepsini topla satır sayısına böl. formül kullanılır

$$J(\theta) = \frac{-\log(\sigma(3)) - \log(\sigma(-1)) - \log(1-\sigma(-5))}{3} \simeq 0.46$$

($Y=1$, $Y=0$)

f) $\frac{\partial J(\theta_1)}{\partial \theta_1}$ ... türev (gradient) $\Rightarrow$ Gradient = (Tahmin − Gerçek değer) . X

(Y'nin o satırdaki değeri, X değeri)

(first row → soruda vermiş)

$(\sigma(z) - Y) \cdot X_i$

1. column için

$(\sigma(3) - 1) \cdot 1 = -0,05$

$\sigma(3)$ verilmiş verilmediyse → $\sigma(z) = \frac{1}{1+e^{-z}}$

2. column

$(\sigma(3) - 1) \cdot 2 = -0,10$

3. column

$(\sigma(3) - 1) \cdot 1 = -0,05$

g) second row $\quad \sigma(-1) \approx 0,25$

1. column.

$((\sigma(-1) - 1) \cdot 1 = -0,75$

2. col. $\quad (\sigma(-1) - 1) \cdot 1 = -0,75$

3. col. $\quad (\sigma(-1) - 1) \cdot (-1) = 0,75$

h) update θ (ağırlık) * Stochastic gradient descent

Yeni ağırlık = Eski ağırlık − (Learning_rate x Grad)

$\theta_{new} = \theta_{old} - (\text{Learn. rate} \times \text{gradient})$

second training demiş (2. satır için)

$$\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} - 1. \begin{bmatrix} -0,75 \\ -0,75 \\ 0,75 \end{bmatrix} = \begin{bmatrix} -1,25 \\ 2,75 \\ 0,25 \end{bmatrix}$$

θ_{old} — learn. rate x grad. — θ_{new}

i) e partindakini göz tekrar

$J(\theta_{new}) = 0,09$. Ancak bu terz hatayı düşürme artırma sonları her daim azalmalı olur. Machine learn. amacı budur. Ama Jik kontrol et.

j) $\theta_{new} = \begin{bmatrix} -1,25 \\ 2,75 \\ 0,25 \end{bmatrix}$ $\quad X = [1, 3, 4]$ vermiş

$z = (-1,25 \cdot 1) + (2,75 \cdot 3) + (0,25 \cdot 4) = 8$

$\sigma(z) = \frac{1}{1+e^{-z}} = 0,99$

$\sigma(8) > 0,5 \Rightarrow Y = 1$

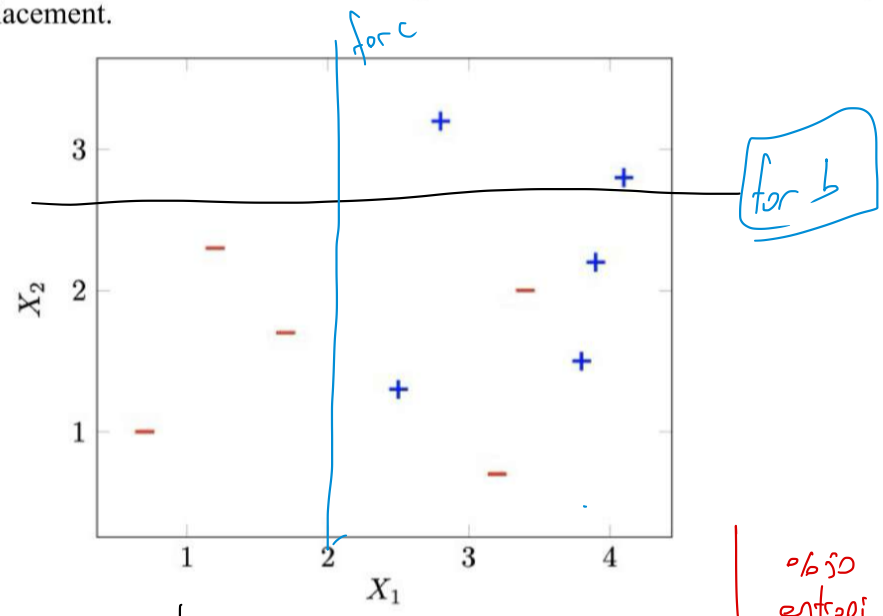
aynı şeydir.

$z > 0$ veya $\sigma > 0,5 \Rightarrow Y = 1$

$z < 0$ veya $\sigma < 0,5 \Rightarrow Y = 0$

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%50 %50 eşit olu. entropi her zaman 1 bit çıkar

a) 5 −
5 +

entropi formülü

$$H(y) = -\left(P(y-)\log_2\left(P(y=-)\right) + \left(P(y+)\log_2 P(y=+)\right)\right)$$

$$= -\left(\frac{1}{2}\log_2(1/2) + \frac{1}{2}\log_2(1/2)\right) = 1$$

b) $X_2 < 2{,}5$ Mutual information => (old Entropy − New entropy)

0,2 ağırlık
2/10 → $X_2 > 2{,}5$ => 2 tane + var %100 saf Entropy = 0

$X_2 < 2{,}5$ => 5 −
3 +

8 tane var
8/10 = 0,8 ağırlıklı

Entropy = $H(alt) = -\left(\frac{5}{8}\log_2\left(\frac{5}{8}\right) + \frac{3}{8}\log_2\left(\frac{3}{8}\right)\right)$

(eski)

$$1 - \left[H(alt)\cdot 0{,}8 + (0 \cdot 0{,}2)\right] = 0{,}2364\overline{5}$$

(c) Hata Oranına (Error Rate) Göre En İyi Bölme Neresidir?

Hoca bu şıkta "Entropi/logaritma hesaplamayı bırak, sadece gözünle bakıp en az hata yaptıran çizgiyi bul" diyor.

- **Hata oranı nedir?** Çizgiyi çektiğinde, her kutuyu içerideki çoğunluğun rengine boyarsın. Azınlıkta kalıp yanlış kutuya düşen noktalar senin "Hatan" olur.
- Grafiğe bakıp x-ekseninden (X_1) **2** hizasında dikey bir çizgi çek ($X_1 < 2$).
- **Sol Taraf ($X_1 < 2$):** 3 tane Eksi (-) var, hiç Artı yok. Kutu saf, hata = 0.
- **Sağ Taraf ($X_1 \geq 2$):** 7 nokta var. 5'i Artı, 2'si Eksi. Çoğunluk Artı olduğu için makine burayı komple "Artı" ilan eder. Arada kalan o 2 tane Eksi bizim hatamızdır.
- Toplam 10 noktada sadece 2 yanlış yaptık. Başka hiçbir düz çizgi sana 2'den az hata vermez. Bu yüzden en iyi çizgi $X_1 < 2$ çizgisidir.

(d) Tek Bölmeyle Görülebilecek En Düşük Hata Oranı Nedir?

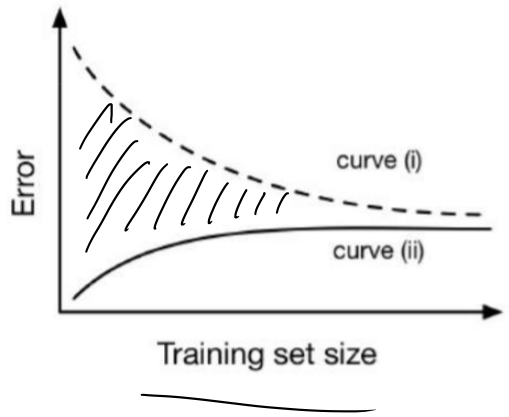
Aslında (c) şıkkında bunu zaten bulduk.

- En iyi çizgiyi çektiğimizde sistemin yaptığı toplam hata **2** tane noktaydı.
- Toplam noktamız 10 olduğuna göre hata oranı: $2/10 = \mathbf{0.2}$
- *(Hocanın cevabındaki ± 0.001 ifadesi muhtemelen küsurat kabul toleransıdır, ana cevap 0.2'dir).*

(e) Sıfır Hata Oranına Ulaşmak Mümkün Mü?

- Hoca soruyor: "Ben o kadar çok çizgi çekeyim ki, herkes kendi sınıfındaki kutuda kalsın ve eğitim hatası 0 olsun, bu mümkün mü?"
- Cevap kesinlikle **Evet (Yes)**'tir. Karar ağaçlarının en bilinen özelliği budur. Ağacı yeterince derinleştirip, x ve y eksenlerinden peş peşe çizgiler çekersen, her bir noktanın etrafını tek tek izole edebilirsin. Bu duruma makine öğrenmesinde **Overfitting (Aşırı Öğrenme / Ezberleme)** denir, test verisinde patlar ama eğitim anında (training error) hatayı sıfıra indirmen her zaman mümkündür.

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(a) [5 pts] Which curve represents the training error? Provide 1-2 sentences of justification.

a) curve ii, Because Eğitim veri seti boyutu arttıkça, modelin tüm noktaları kusursuz bir şekilde ezberlemesi zorlaşır. Bu yüzden training error artar

b) Overfitting

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a) 1) Regression (Regresyon)
2) Classification (Sınıflandırma)
3) Clustering (Kümeleme)
4) Dimensionality Reduction (Boyut azaltma)

b) -> Estimate the price of a house => Regression
-> Email is spam or not => Classification

* Tahmin edilecek şey sayısal (fiyat, sıcaklık) => Regression
" " " etiketse (kedi, köpek, spam) => Classification
Amaç etiketsiz verileri Gruplamaksa (Müşteri tiplerini bulmak) => Clustering
Amaç çok sütunlu veriyi Sıkıştırmaksa (Hız ve özet için) =>